INCH-POUND

MIL-E-85082/3

16 November 1992

MILITARY SPECIFICATION SHEET

ENCODER, SHAFT POSITION TO DIGITAL
OPTICAL INCREMENTAL
TYPE 11SKL05S05A

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Encoders, Shaft Angle to Digital, described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) cited in the solicitation: MIL-E-85082.

TABLE I. Requirements.

| Requirements | Value | Unit | Tolerance |
|---|---|---|---|
| Code and scan | Incremental (w/o Zero) | | |
| Capacity | $2^5$ | counts | maximum |
| Resolution | $2^5$ | counts/turn | |
| Full count revolutions | 1 | turn | |
| Voltage interrogation | +5.0 | volts DC | maximum |
| Starting torque | 0.2 | oz-in | minimum |
| Running torque | 2.0 | oz-in | maximum |
| Operating speed | 500 | rpm | maximum |
| Slew speed | 1500 | rpm | maximum |
| Binary "0" | 0 | volts DC | ±0.5 |
| Binary "1" | 5 | volts DC | ±0.5 |
| Perpendicularity | 0.0010 | inch | maximum |
| Operating temperature | -55° to +85° | degrees C | min-max |
| Transition accuracy | ±1/2 | bit | |
| Moment of inertia | 1.6 | $oz\text{-}in^2$ | maximum |
| Weight (without leads) | 0.4 | lbs | maximum |
| Life | 10,000,000 | revolutions | |

AMSC - N/A FSC 5990

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TABLE II. Part identifying number variant characteristics.

| Part Identifying No. 1/ | A ±0.010 | E Maximum | G Maximum | Shaft Style | Terminal End | U +0.0000 −0.0002 |
|---|---|---|---|---|---|---|
| M85082/3-01A | .550 | .6250 | 2.000 | Plain | Wire Leads | .1248 |

1/ Part identifying number M85082/3-01A shall be the NATO standard.

1/ The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation.

TABLE III. Wire termination identification.

| FUNCTION | COLOR |
|---|---|
| +5VDC | RED/VIOLET |
| GROUND | BLACK |
| OUTPUT A | WHITE/BROWN |
| OUTPUT B | WHITE/RED |

NOTES:

1. Dimension C shall be 1.062 with a tolerance of +0.000/-0.003.
2. Dimension L shall be 0.975 with a tolerance of +0.005/-0.000.
3. Shaft may be sealed against moisture and electromagnetic interference with a conducting type "O" ring (see Figure 1).
4. Rotation for increasing count shall be clockwise.
5. Wire leads shall be in accordance with MIL-W-16878/4, size 26 and shall be a minimum of 15 inches long.



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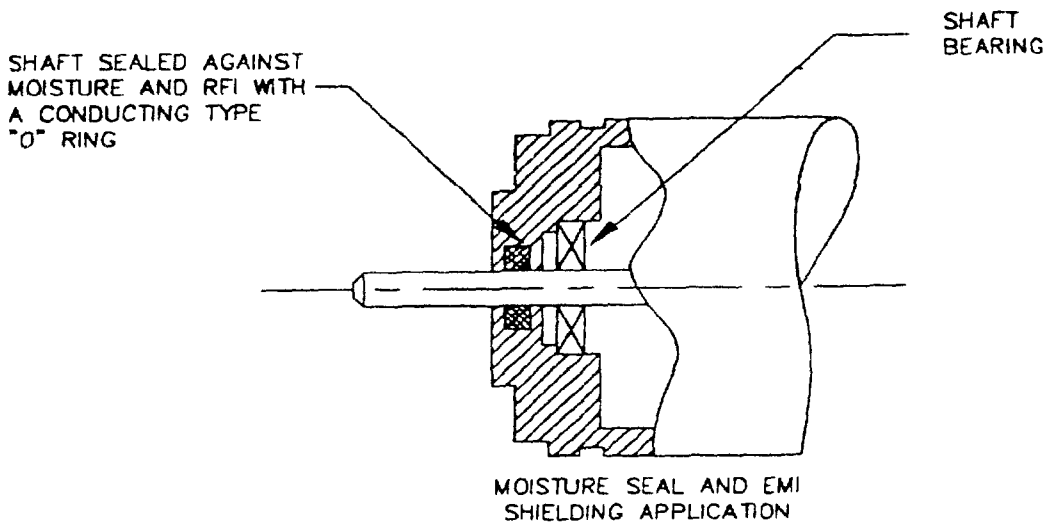



FIGURE 1 Moisture seal and EMI shielding application.

Custodians:
Navy - AS
Army - AR
Air Force - 85

Review activities:
DLA - ES

Preparing activity:
Navy - AS

(Project 5990-0405-3)



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